

RaceDay API Endpoint Plan

Table 1: Authentication

HTTP method	Route	Description	Role required	Request body	Expected response
POST	/api/v1/auth/register	Register a new user (Participant or Organiser).	None (Public)	{ "email": "string", "password": "string", "fullName": "string", "role": "Participant Organiser", "phone": "string", "dateOfBirth": "2026-09-03" (Participant only) }	201 Created – { "userID": 1, "email": "...", "role": "...", "fullName": "..."} 400 Bad Request – duplicate email or validation error
POST	/api/v1/auth/login	Authenticate and receive JWT token.	None (Public)	{ "email": "string", "password": "string" }	200 OK – { "token": "jwt_string", "expiresIn": 86400 } 401

					Unauthorized – invalid credentials
POST	/api/v1/auth/refresh	Refresh an expired JWT.	None (Public)	{ "refreshToken": "string" }	200 OK – { "token": : "jwt_string", "expiresIn": 86400 } 401 Unauthorized – invalid refresh token

Table 2: User Profile

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/v1/users/me	Get the current user's profile (includes role-specific fields).	Authenticated (Any)	None	200 OK – { "userID": 1, "email": "...", "role": "Participant Organiser", "fullName": "...", "phone": "...", "firstName": "...", "lastName":

					"...", "dateOfBirth": "..." } 401 Unauthorized
PUT	/api/v1/users/me	Update the current user's own profile.	Authenticated (Any)	{ "fullName": "string", "phone": "string", "emergencyContact": "string" } (fields optional)	200 OK – Updated profile object 400 Bad Request 404 Not Found

Table 3: Organisers

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/v1/organisers	List all organisers.	Organiser / Admin	None	200 OK – [{ "organiserID": 1, "userID": 1, "name": "...", "address": "..."}] 401 Unauthorized
GET	/api/v1/organisers/{id}	Get a specific organiser by ID.	Organiser / Admin	None	200 OK – Single organiser object 404 Not Found

POST	/api/v1/organisers	Create an organiser profile for an existing User.	None (Public)	{ "userID": 1, "name": "string", "address": "string" }	201 Created – Returns new organiser with organiserID 400 Bad Request (userID already has organiser)
PUT	/api/v1/organisers/{id}	Update an organiser's details.	Organiser (Self)	{ "name": "string", "address": "string" } (Partial)	200 OK – Updated organiser 403 Forbidden / 404 Not Found
DELETE	/api/v1/organisers/{id}	Delete an organiser (checks for existing events).	Admin only	None	204 No Content – Success 409 Conflict (has existing events)

Table 4: Participants

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/v1/participants	List all participants.	Organiser / Admin	None	200 OK – [{ "participantID": 1, "userID": 5, "firstName":

					<pre>"...", "lastName": "...", "dateOfBirth": "...", "gender": "...", "emergencyContact": "..." }]</pre>
GET	/api/v1/participants/{id}	Get a specific participant by ID.	Self or Organizer	None	200 OK – Single participant object 404 Not Found
POST	/api/v1/participants	Create a participant profile for an existing User.	None (Public)	<pre>{ "userID": 5, "firstName": "string", "lastName": "string", "dateOfBirth": "2026-09-03", "gender": "M</pre>	<pre>F", "emergencyContact": "string" }</pre>
PUT	/api/v1/participants/{id}	Update a participant's profile.	Self or Organizer	<pre>{ "firstName": "string", "lastName": "string", "emergencyContact": "string" } (Partial)</pre>	200 OK – Updated participant 403 Forbidden / 404 Not Found
DELETE	/api/v1/participants/{id}	Delete a participant	Admin only	None	204 No Content – Success 409 Conflict

		(checks for active entries).			(has existing entries)
--	--	------------------------------	--	--	------------------------

Table 5: Events

HTT P me tho d	Route	Description	Role requ ired	Request body	Expected response
GET	/api/v1/events	List all events (filters: ?status=Open, ?location=Cape%20Town) .	Non e (Pub lic)	None	200 OK – [{ "event ID": 1, "organise rID": 1, "name": "...", "eventDat e": "...", "location ": "...", "status": "..."}]
GET	/api/v1/events/upcoming	List events with date ≥ today, sorted ascending.	Non e (Pub lic)	None	200 OK – Array of events (same structure as above)

GET	/api/v1/events/{id}	Get full event details (includes organiser, categories, weather, and route).	None (Public)	None	200 OK – Full event object with organiser, categories, and all weather/route fields 404 Not Found
POST	/api/v1/events	Create a new event.	Organiser	{ "organiserID": 1, "name": "string", "description": "string", "eventDate": "2026-11-29", "startTime": "06:00", "location": "string", "status": "Open" }	201 Created – Returns new event with eventID 400 Bad Request / 403 Forbidden
PUT	/api/v1/events/{id}	Update an existing event (including weather & route fields).	Organiser (Owner)	{ "name": "...", "description": "...",	200 OK – Updated event 403 Forbidden /

				"eventDate": "...", "startTime": "...", "location": "...", "status": "...", "startPoint": "...", "finishPoint": "...", "mapUrl": "...", "elevationUrl": "...", "weatherForecast": "...", "temperature": 22.5, "windSpeed": 10, "precipitation": 0 } (all optional)	404 Not Found
DELETE	/api/v1/events/{id}	Delete an event.	Organiser (Owner)	None	204 No Content – Success 409

					Conflict (has entries)
GET	/api/v1/events/{id}/weather	Convenience: Get only weather data for an event.	None (Public)	None	200 OK – { "eventId": 1, "forecastDate": "2026-11-29", "weatherForecast": "Mild", "temperature": 17.0, "windSpeed": 15, "precipitation": 20 } 404 Not Found
GET	/api/v1/events/{id}/route	Convenience: Get only route data for an event.	None (Public)	None	200 OK – { "eventId": 1, "startPoint": "Durban City Hall", "finishPoint": "Pietermaritzburg Oval",

					"mapUrl": "https://... , "elevationUrl": "https://... , "distanceKm": 87.0 } 404 Not Found
--	--	--	--	--	---

Table 6: Categories

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/v1/events/{eventId}/categories	Get all categories for an event.	None (Public)	None	200 OK – [{"categoryID": 1, "eventID": 1, "name": "...", "description": "...", "minAge": 20, "maxAge": 39, "gender": "M", "distance": 87.0 }]

					404 Not Found
GET	/api/v1/categories/{id}	Get a single category by ID.	None (Public)	None	200 OK – Single category object 404 Not Found
POST	/api/v1/events/{eventId}/categories	Add a new category to an event.	Organiser (Owner)	{ "name": "string", "description": "string", "minAge": 0, "maxAge": 99, "gender": "M"	F
PUT	/api/v1/categories/{id}	Update a category.	Organiser (Owner)	Same as POST (fields optional)	200 OK – Updated category 403 Forbidden / 404 Not Found
DELETE	/api/v1/categories/{id}	Delete a category (checks for entries).	Organiser (Owner)	None	204 No Content – Success 409 Conflict (has existing entries)

Table 7: Entries

HTT P met hod	Route	Descrip tion	Role requi red	Request body	Expected response
GET	/api/v1/participants/{participantId}/entries	Get all entries for a specific participant.	Self or Organiser	None	200 OK – [{"entryID": 1, "participantID": 1, "eventID": 1, "categoryID": 1, "registrationDate": "...", "status": "...", "startNumber": 101, "paid": 1 }]
GET	/api/v1/events/{eventId}/entries	Get all entries for a specific event (organiser dashboard).	Organiser (Owner)	None	200 OK – Entries array with participant details 404 Not Found
GET	/api/v1/entries/{id}	Get a specific	Self or	None	200 OK – Single entry

		entry by ID.	Organiser		object 404 Not Found
POST	/api/v1/entries	Register a participant for an event/category.	Participant (Self)	{ "participantID": 1, "eventID": 1, "categoryID": 1, "status": "Pending", "paid": 0 }	201 Created – Returns new entry with entryID 400 Duplicate / 403 Forbidden
PUT	/api/v1/entries/{id}	Update an entry (organisers update status/start number; participants update paid flag).	Organiser or Self	{ "status": "Confirmed", "startNumber": 101, "paid": 1 } (Partial)	200 OK – Updated entry 403 Forbidden / 404 Not Found
DELETE	/api/v1/entries/{id}	Withdraw an entry (cancel registration).	Self or Organiser	None	204 No Content – Success 409 Conflict (race started / result exists)

Table 8: Results

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/v1/entries/{entryId}/result	Get the result for a specific entry.	Self or Organizer	None	200 OK – { "resultID": 1, "entryId": 1, "finishTime": "08:45:30", "position": 523, "status": "Finished", "chipTime": "08:42:15", "gunTime": "08:45:30" } 404 No result yet
GET	/api/v1/events/{eventId}/results	Get all results for an event (ranked leaderboard).	None (Public)	None	200 OK – Array of results with participant names and positions

					404 Not Found
POST	/api/v1/entries/{entryId}/result	Upload a result for a finished entry.	Organiser (Owner)	{ "finishTime": "08:45:30", , "position": 523, "status": "Finished", , "chipTime": : "08:42:15", , "gunTime": "08:45:30" }	201 Created – Returns new result object 400 Bad format / 409 Conflict (result exists)
PUT	/api/v1/results/{id}	Update an existing result.	Organiser (Owner)	Same as POST (fields optional)	200 OK – Updated result 404 Not Found
DELETE	/api/v1/results/{id}	Delete a result.	Organiser (Owner)	None	204 No Content – Success 404 Not Found

Summary of HTTP Status Codes

Code	Meaning	When used
200 OK	Success	GET / PUT requests

201 Created	Resource created	POST requests
204 No Content	Deleted	DELETE requests
400 Bad Request	Client error	Invalid JSON, missing fields, duplicate data
401 Unauthorized	Not logged in	Missing / invalid JWT
403 Forbidden	Insufficient permissions	Wrong role or not resource owner
404 Not Found	Resource missing	ID does not exist
409 Conflict	Business rule violation	Dependent records exist, duplicate entry